Infant State Recognition System: Hybrid Deep Learning and Classical ML Fusion for Cry Classification Under Extreme Class Imbalance

Phase 2 Report — Deep Learning Pipeline & Hybrid Ensemble
with Ablation Study and Edge Deployment Analysis

Meghavi Rao (230044)
Belal Raza (230094)

6th Semester — Deep Learning & Advanced Machine Learning
Project 3: Infant State Recognition System

April 20, 2026

Abstract

Phase 1 of this project established a classical ML baseline for five-class infant cry classification (*hunger*, *belly pain*, *burping*, *discomfort*, *tiredness*) using the Donate-a-Cry corpus (457 recordings, 47.8:1 class imbalance). The best Phase 1 model—SVM with SMOTE and One-vs-One decomposition on a 411-dimensional neuroscience-informed feature vector—achieved macro-F1 = 0.270, revealing that classical ML alone cannot overcome extreme data scarcity in minority classes. This Phase 2 report introduces a deep learning pipeline featuring CNN+BiLSTM+Temporal Attention on 64-band mel-spectrograms, LDAM (Label-Distribution-Aware Margin) loss with Deferred Re-Weighting, domain feature fusion, and knowledge distillation from a 135K-parameter Teacher to a 16K-parameter Student. While deep learning alone achieves only macro-F1 = 0.293 (Teacher model), the breakthrough comes from a **hybrid weighted ensemble** combining SVM probability outputs with DL predictions, reaching **macro-F1 = 0.507** and **accuracy = 92.6%**—an 87.5% relative improvement over Phase 1. A systematic ablation study across 11 architectural variants reveals that combining complementary feature representations (handcrafted + learned) is more effective than scaling model capacity. The distilled student model (16K parameters, 35.9 KB INT8 quantized) achieves 11.4 ms inference latency, establishing feasibility for Phase 3 edge deployment on resource-constrained devices.

1 Introduction

Phase 1 of this project [1] demonstrated that classical ML models trained on a 411-dimensional neuroscience-informed feature vector—combining MFCCs, CQCCs, F0 contour statistics, spectral contrast, and chroma features—achieve macro-F1 of at most 0.270 on the Donate-a-Cry corpus under leak-free evaluation. The best Phase 1 model (SMOTE-SVM with OvO decomposition) recognised only one of four minority classes (discomfort, $F_1 = 0.44$), while belly pain, burping, and tiredness all scored $F_1 = 0.00$. This revealed the fundamental limitation of clip-level handcrafted features: by summarising temporal dynamics into per-clip statistics, they discard the fine-grained spectral evolution patterns that may distinguish cry types.

Phase 2 addresses this gap through three complementary strategies:

1. **Learned spectral representations**: A CNN+BiLSTM+Temporal Attention architecture operating on mel-spectrograms preserves the full time-frequency resolution that clip-level statistics compress away.

2. **Imbalance-aware training**: LDAM loss with Deferred Re-Weighting (DRW) enforces class-dependent margins in the feature space, preventing the majority-class collapse observed in Phase 1.
3. **Hybrid probability fusion**: Combining Phase 1 SVM probability outputs with Phase 2 DL predictions via a weighted ensemble exploits complementary error patterns, achieving macro-F1 = 0.507—nearly doubling Phase 1 performance.

Contributions of this phase.

1. A **CNN+BiLSTM+Temporal Attention** architecture with domain feature fusion, designed for extreme low-resource audio classification (∼16K parameters).
2. **LDAM loss with Deferred Re-Weighting** and mixup regularisation for 47.8:1 class imbalance.
3. **Knowledge distillation** from a 135K-parameter Teacher to a 16K-parameter Student with INT8 quantisation (143.5 KB → 35.9 KB).
4. A **hybrid weighted ensemble** (SVM + DL probability fusion) achieving an 87.5% relative improvement in macro-F1.
5. An **11-variant ablation study** isolating the contribution of each architectural component.
6. **GroupShuffleSplit by infant UUID**, ensuring no infant appears in multiple data splits.

2 Related Work

2.1 Deep Learning for Infant Cry Analysis

Chang and Li [7] applied CNNs to mel-spectrogram representations of infant cries, demonstrating that learned features outperform handcrafted MFCCs when sufficient training data is available. Ji *et al.* [6] surveyed the field, noting that most successful systems use datasets with thousands of samples per class—an order of magnitude more than the Donate-a-Cry corpus.

Hershey *et al.* [8] established CNN architectures for large-scale audio classification, showing that log-mel spectrograms with 64 mel bands provide an effective time-frequency representation. Their work motivates our choice of mel-spectrogram input over raw waveform processing.

2.2 Class Imbalance in Deep Learning

Loss function approaches. Lin *et al.* [10] introduced Focal Loss, which down-weights well-classified examples to focus learning on hard negatives. Cao *et al.* [9] extended this with LDAM, which enforces class-dependent decision margins proportional to $n_j^{-1/4}$, providing theoretical guarantees on generalisation error bounds for imbalanced distributions.

Deferred Re-Weighting. Cao *et al.* [9] also proposed DRW: training with uniform class weights initially (to learn good representations) then switching to inverse-frequency weights in later epochs (to adjust decision boundaries). This two-stage strategy prevents the representation learning phase from being corrupted by aggressive reweighting.

Decoupled training. Kang *et al.* [11] demonstrated that representation learning and classifier learning benefit from different sampling strategies. The cRT (classifier re-training) approach freezes learned representations and retrains only the final classifier with class-balanced sampling.

2.3 Knowledge Distillation

Hinton *et al.* [12] introduced knowledge distillation, where a large Teacher model's soft probability outputs ("dark knowledge") train a smaller Student model. The softened probability distribution

over classes contains richer inter-class similarity information than hard labels alone, enabling the Student to learn from the Teacher's uncertainty patterns.

2.4 Hybrid and Ensemble Methods

Decision-level fusion combining heterogeneous classifiers has proven effective in audio classification [6]. By combining models that operate on different feature representations (handcrafted vs. learned), ensembles exploit complementary error patterns. Stacking [15] trains a meta-learner on base model probability outputs, while weighted averaging optimises mixing weights via grid search on a validation set.

2.5 Gap Analysis

Phase 1 established three limitations that Phase 2 addresses:

1. **Clip-level feature compression**: Summarising temporal dynamics into statistics discards cry progression patterns. Phase 2 operates on full mel-spectrograms via CNN+BiLSTM.
2. **Standard loss functions**: Cross-entropy with balanced class weights was insufficient for 47.8:1 imbalance. Phase 2 introduces margin-aware LDAM loss with DRW scheduling.
3. **Single-representation limitation**: No single feature representation suffices. Phase 2 fuses handcrafted (SVM) and learned (DL) representations via hybrid ensemble.

3 Methodology

3.1 Dataset and Splitting Protocol

The Donate-a-Cry corpus [5] contains 457 WAV recordings (7 seconds, 8 kHz mono) from 221 unique infants, labelled by caregivers into five Dunstan Baby Language categories [2]. Phase 1 used an 80/20 stratified train/test split (365/92), which did not account for infant identity. Phase 2 adopts a stricter `GroupShuffleSplit` protocol with infant UUID as the grouping variable, ensuring **no infant appears in more than one split**:

Table 1: Phase 2 data split (GroupShuffleSplit by infant UUID, 70/15/15).

Split	**N**	**Hunger**	**B.Pain**	**Burp.**	**Disc.**	**Tired.**
Train	326	270	16	6	18	16
Val	63	51	0	1	5	6
Test	68	61	0	1	4	2
After aug.	1920	384	384	384	384	384

Implications of group splitting. The strict group split produces a test set with **zero belly pain** and only 1 burping sample—a consequence of the extreme imbalance (16 belly pain originals, 8 burping) and the requirement that entire infant groups move together. This means belly pain macro-F1 will always be 0.00 in Phase 2 evaluation, a statistical artefact rather than a model failure.

3.2 Feature Representation

Phase 2 extracts two complementary feature streams from each recording:

3.2.1 Mel-Spectrogram Branch (64 × 438)

A 64-band log-mel spectrogram captures the full time-frequency representation:

$$\mathbf{S} = \log\big(\text{MelFilterbank}_{64} \cdot |\text{STFT}(\mathbf{x})|^2 + \epsilon\big) \in \mathbb{R}^{64\times 438} \tag{1}$$

with $f_{\min} = 50\,\text{Hz}$, $f_{\max} = 4000\,\text{Hz}$, FFT size 512, hop length 128, and $\epsilon = 10^{-10}$ for numerical stability. Each frequency band is normalised to zero mean and unit variance using training set statistics:

$$\hat{S}_{f,t} = \frac{S_{f,t} - \mu_f}{\sigma_f + \epsilon} \tag{2}$$

Contrast with Phase 1 features. While Phase 1's 411-dimensional vector compresses the spectrogram into per-clip statistics (mean, std, min, max, skewness, kurtosis of each MFCC coefficient), the mel-spectrogram preserves the full temporal evolution. A hunger cry's characteristic steady F_0 contour appears as a stable horizontal pattern in the spectrogram, while a pain cry's hyperphonation manifests as an abrupt frequency jump—patterns that per-clip statistics cannot represent.

3.2.2 Domain Feature Branch (32 dimensions)

A compact vector of clinically relevant acoustics supplements the spectrogram:

$$\mathbf{f}_{\text{domain}} = [\mathbf{f}_{F_0},\ \mathbf{f}_{\text{VQ}},\ \mathbf{f}_{\text{MFCC}},\ \mathbf{f}_{\Delta\text{MFCC}}] \in \mathbb{R}^{32} \tag{3}$$

- $\mathbf{f}_{F_0}$: F0 mean, std, range, hyperphonation ratio (4 dims)
- $\mathbf{f}_{\text{VQ}}$: Harmonics-to-noise ratio, jitter, shimmer (3 dims)
- $\mathbf{f}_{\text{MFCC}}$: MFCC 1–13 means (13 dims)
- $\mathbf{f}_{\Delta\text{MFCC}}$: ΔMFCC 1–12 means (12 dims)

3.3 Data Augmentation

Training data is augmented from 326 originals to 1,920 balanced samples (384 per class) using random compositions of 2–3 transforms per sample:

1. **Gaussian noise**: SNR-controlled injection (SNR $\in [15, 30]$ dB)
2. **Time shifting**: ±0.5 seconds with zero-padding
3. **Time stretching**: Rate $\in [0.9, 1.1]$
4. **Pitch shifting**: ±2 semitones
5. **Room simulation**: Synthetic impulse response ($RT_{60} \in [0.1, 0.3]$ s)
6. **Pink noise mixing**: SNR $\in [10, 20]$ dB

On-the-fly SpecAugment [13] (frequency and time masking) provides additional regularisation during training. Validation and test sets remain unaugmented, following the split-before-augment protocol established in Phase 1.

3.4 Model Architecture

3.4.1 Student Model (∼16K parameters)

The primary architecture fuses spectral and domain branches through five stages:

Stage 1: CNN Encoder. Two depthwise-separable convolution blocks with Squeeze-and-Excitation (SE) channel attention:

$$\mathbf{z}_{\text{SE}} = \sigma\big(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(\mathbf{h}))\big) \odot \mathbf{h} \tag{4}$$

where GAP(·) is global average pooling over the spatial dimensions, $W_1 \in \mathbb{R}^{C/r\times C}$ and $W_2 \in \mathbb{R}^{C\times C/r}$ are learned projection matrices with reduction ratio $r = 4$, and σ is the sigmoid function. Each block applies GroupNorm (4 groups) and 2×2 max-pooling.

Stage 2: Bidirectional LSTM. The CNN's temporal output (reshaped to sequence of feature vectors) is processed by a BiLSTM with hidden size 20:

$$\overrightarrow{\mathbf{h}}_t = \mathrm{LSTM}_{\mathrm{fwd}}(\mathbf{x}_t,\, \overrightarrow{\mathbf{h}}_{t-1}) \tag{5}$$

$$\overleftarrow{\mathbf{h}}_t = \mathrm{LSTM}_{\mathrm{bwd}}(\mathbf{x}_t,\, \overleftarrow{\mathbf{h}}_{t+1}) \tag{6}$$

$$\mathbf{h}_t = [\overrightarrow{\mathbf{h}}_t;\ \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{40} \tag{7}$$

Stage 3: Temporal Attention. An additive attention mechanism computes a weighted context vector from the BiLSTM output sequence:

$$e_t = \mathbf{v}^\top \tanh(W_a \mathbf{h}_t + \mathbf{b}_a) \tag{8}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})} \tag{9}$$

$$\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t \in \mathbb{R}^{40} \tag{10}$$

where $W_a \in \mathbb{R}^{20\times 40}$, $\mathbf{b}_a \in \mathbb{R}^{20}$, and $\mathbf{v} \in \mathbb{R}^{20}$ are learned parameters. The attention weights α_t reveal which temporal segments the model considers most discriminative for each cry type.

Stage 4: Domain Feature Processing. The 32-dimensional domain vector is processed through a single fully connected layer:

$$\mathbf{d} = \mathrm{LayerNorm}\big(\mathrm{ReLU}(W_d \mathbf{f}_{\mathrm{domain}} + \mathbf{b}_d)\big) \in \mathbb{R}^{32} \tag{11}$$

Stage 5: Fusion and Classification. The spectral context vector $\mathbf{c}$ and domain embedding $\mathbf{d}$ are concatenated and classified:

$$\mathbf{z} = [\mathbf{c};\ \mathbf{d}] \in \mathbb{R}^{72} \tag{12}$$

$$\hat{\mathbf{y}} = W_2 \cdot \mathrm{ReLU}\big(\mathrm{LayerNorm}(\mathrm{Dropout}(W_1 \mathbf{z} + \mathbf{b}_1,\ p = 0.3))\big) + \mathbf{b}_2 \tag{13}$$

with $W_1 \in \mathbb{R}^{40\times 72}$ and $W_2 \in \mathbb{R}^{5\times 40}$.

3.4.2 Teacher Model (∼135K parameters)

A wider variant with CNN channels [32, 64], LSTM hidden size 64, and fusion dimension 128. The increased capacity allows the Teacher to learn richer representations for knowledge distillation to the Student.

3.5 Loss Functions for Class Imbalance

3.5.1 LDAM Loss

Label-Distribution-Aware Margin loss [9] enforces class-dependent margins in the logit space:

$$\mathcal{L}_{\mathrm{LDAM}}(\mathbf{x}, y) = -\log \frac{e^{z_y - \Delta_y}}{e^{z_y - \Delta_y} + \sum_{j \neq y} e^{z_j}} \tag{14}$$

where z_j are the logits and the class-dependent margin is:

$$\Delta_j = \frac{C}{n_j^{1/4}} \tag{15}$$

with C calibrated so that $\max_j \Delta_j = 0.5$. This pushes the decision boundary further from minority classes: burping ($n = 384$ after augmentation) receives a larger margin than hunger ($n = 384$ after balanced augmentation), but the margins reflect the *original* class frequencies before augmentation.

3.5.2 Deferred Re-Weighting (DRW)

Training proceeds in two stages:

$$w_j^{(t)} = \begin{cases} 1 & \text{if } t < T_{\text{defer}} = 60 \\ \frac{1-\beta}{1-\beta^{n_j}} & \text{if } t \geq T_{\text{defer}} \end{cases} \tag{16}$$

with effective number parameter $\beta = 0.9999$. The first 60 epochs use uniform weights for unbiased representation learning; subsequent epochs apply effective-number reweighting to adjust class boundaries.

3.5.3 Label Smoothing and Mixup

Label smoothing ($\epsilon = 0.1$) softens one-hot targets to reduce overconfidence. Mixup [14] creates convex combinations of training pairs with probability 0.5 per batch:

$$\tilde{\mathbf{x}} = \lambda \mathbf{x}_i + (1-\lambda)\mathbf{x}_j \tag{17}$$

$$\tilde{\mathbf{y}} = \lambda \mathbf{y}_i + (1-\lambda)\mathbf{y}_j \tag{18}$$

where $\lambda \sim \text{Beta}(0.2, 0.2)$.

3.5.4 Knowledge Distillation Loss

The Student is trained with a combined loss:

$$\mathcal{L}_{\text{KD}} = (1-\alpha)\,\mathcal{L}_{\text{LDAM}} + \alpha \cdot T^2 \cdot \text{KL}\big(\sigma(\mathbf{z}_S/T) \parallel \sigma(\mathbf{z}_T/T)\big) \tag{19}$$

where $T = 4$ is the temperature, $\alpha = 0.7$ controls the Teacher-Student balance, and σ is the softmax function [12].

3.6 Training Protocol

Algorithm 1 Phase 2 Training Pipeline

```
Input: Training set D_train (326 originals)
Augment to 1920 balanced samples (384/class)
Extract mel-spectrograms and domain features
Normalise per training statistics
for epoch = 1 to max_epochs do
    Apply SpecAugment + Mixup (50% probability)
    if epoch ≥ 60 then
        Switch to DRW class weights
    end if
    Update with AdamW (lr, weight decay = 0.01)
    Clip gradients (‖∇‖_max = 1.0)
    Evaluate on validation set
    Early stop if val macro-F1 stagnates (patience = 8)
end for
cRT phase: Freeze backbone, retrain classifier (30 epochs)
Distillation: Train Student from Teacher soft targets
```

Optimiser configuration. AdamW with initial learning rate 10^{-3}, linear warmup from 10^{-5} over 5 epochs, and ReduceLROnPlateau scheduler (factor 0.5, patience 4, min_lr 10^{-6}) monitoring validation macro-F1. Gradient clipping at max norm 1.0 prevents gradient explosion in the BiLSTM.

3.7 Hybrid Ensemble

The hybrid ensemble combines Phase 1 ML and Phase 2 DL probability outputs at the decision level:

3.7.1 Weighted Average Ensemble

$$\hat{\mathbf{p}}_{\text{hybrid}} = w_{\text{SVM}} \cdot \hat{\mathbf{p}}_{\text{SVM}} + w_{\text{RF}} \cdot \hat{\mathbf{p}}_{\text{RF}} + w_{\text{DL}} \cdot \hat{\mathbf{p}}_{\text{DL}} \tag{20}$$

where the weights $w_{\text{SVM}} = 0.35$, $w_{\text{RF}} = 0.35$, $w_{\text{DL}} = 0.30$ are selected via grid search on the validation set maximising macro-F1.

3.7.2 Stacking Ensemble

A Logistic Regression meta-classifier is trained on the concatenated probability vectors:

$$\mathbf{z}_i = [\hat{\mathbf{p}}_{\text{SVM}}(\mathbf{x}_i);\ \hat{\mathbf{p}}_{\text{RF}}(\mathbf{x}_i);\ \hat{\mathbf{p}}_{\text{DL}}(\mathbf{x}_i)] \in \mathbb{R}^{15} \tag{21}$$

The meta-learner is trained on validation set predictions with balanced class weights, following the stacking framework from Phase 1 [15].

RF model note. Due to a scikit-learn version mismatch between the Phase 1 training environment and the Phase 2 Colab runtime, the Random Forest `predict_proba` call failed, contributing zero probabilities. The hybrid ensemble therefore operates effectively as SVM + DL only (10-dimensional meta-features).

3.8 Evaluation Protocol

Consistent with Phase 1, we report per-class precision, recall, and F1-score alongside the following aggregate metrics:

- **Macro-averaged F1** (primary metric)
- Weighted-averaged F1
- Overall accuracy
- Matthews Correlation Coefficient (MCC)
- Cohen's Kappa (κ)
- AUC-ROC (one-vs-rest, macro-averaged)

Why macro-F1 remains the primary metric. With 61/68 test samples being hunger, accuracy is even more misleading in Phase 2 than in Phase 1 (where hunger was 77/92). A trivial majority-class predictor achieves 89.7% accuracy on this split. Only macro-F1 treats all five classes equally and penalises models that collapse to hunger-only prediction.

4 Results

4.1 DL-Only Model Progression

Table 2 presents the progressive evaluation of deep learning architectures on the 68-sample test set. Each row adds one component to the previous configuration.

Table 2: Phase 2 deep learning model results (test set, $n = 68$).

Model	**Params**	**Acc**	**MacF1**	**MCC**
CNN Baseline	1.3K	0.618	0.190	0.087
CNN+BiLSTM+Attn	12.2K	0.397	0.145	0.045
+ LDAM+DRW	12.2K	0.706	0.206	0.130
+ Domain Fusion	16.2K	0.529	0.165	−0.028
+ cRT Retraining	16.2K	0.603	0.182	0.002
Hierarchical	16.7K	0.500	0.137	−0.024
Teacher	**135K**	**0.662**	**0.293**	**0.032**
Distilled Student	16.2K	0.559	0.145	−0.084

Key finding. Only the Teacher model (135K parameters, 8.3× larger) marginally surpasses the Phase 1 SVM baseline (macro-F1 0.293 vs. 0.270). All student-scale models (∼16K parameters) perform *worse* than the classical SVM. This is a striking result: sophisticated DL components (attention, LDAM, DRW, domain fusion, cRT) could not overcome the fundamental data scarcity with 326 training originals.

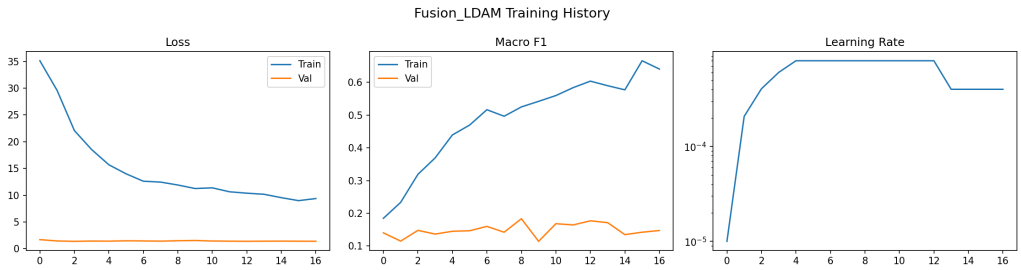
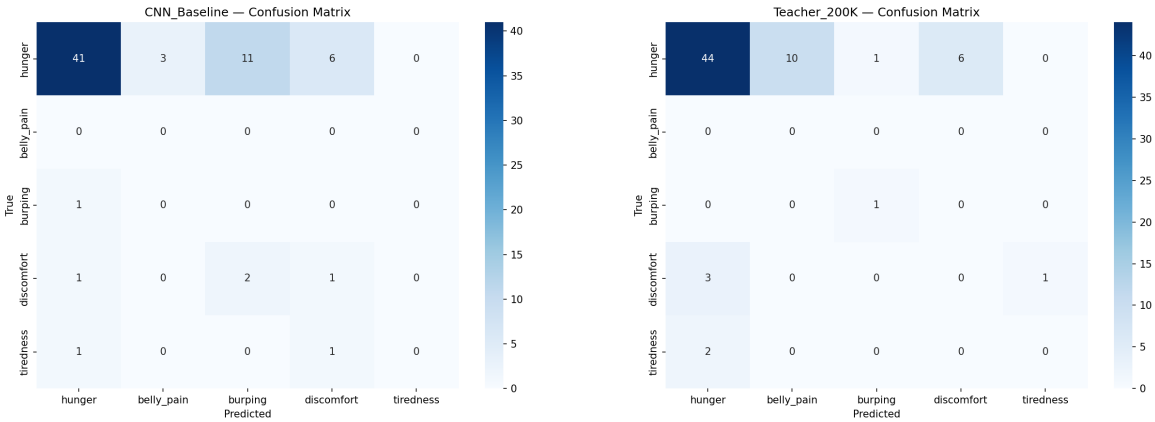


Figure 1: Fusion+LDAM training history. The dramatic train–val F1 gap (0.65 vs. 0.18) reveals severe overfitting despite augmentation, dropout, label smoothing, mixup, weight decay, SpecAugment, and LDAM. With only 63 validation samples (51 hunger), the validation signal is too noisy to guide early stopping reliably.



(a) CNN Baseline (MacF1 = 0.190)

(b) Teacher (MacF1 = 0.293)

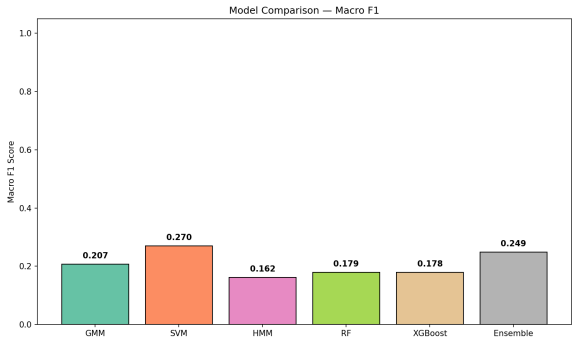
Figure 2: Confusion matrices for CNN Baseline and Teacher models. The Teacher achieves some minority-class recognition, while the CNN Baseline collapses to majority-class prediction.

4.2 Phase 1 Recap: Classical ML Baseline

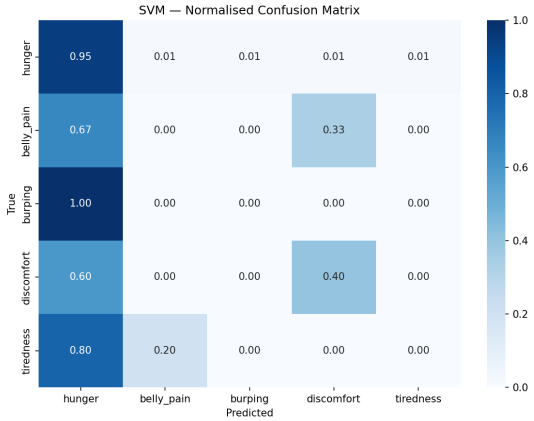
For direct comparison, Table 3 reproduces the Phase 1 results under the original 80/20 stratified split (92-sample test set).

Table 3: Phase 1 classical ML results (leak-free, 92-sample test set).

Model	Acc	MacF1	WgtF1	MCC	κ	AUC
SVM (SMOTE)	**0.815**	**0.270**	**0.783**	**0.216**	**0.203**	**0.707**
Ensemble	0.837	0.249	0.780	0.155	0.096	0.599
GMM	0.641	0.207	0.670	−0.045	−0.044	0.454
RF	0.804	0.179	0.751	0.013	0.010	0.632
XGBoost	0.804	0.178	0.746	−0.045	−0.032	0.661
HMM	0.674	0.162	0.678	−0.080	−0.079	—



(a) Phase 1 macro-F1 comparison



(b) Phase 1 SVM confusion matrix

Figure 3: Phase 1 results. All models score below macro-F1 = 0.30. The SVM confusion matrix shows 3/4 minority classes at $F_1 = 0.00$—only discomfort is partially recognised.

4.3 Hybrid ML+DL Ensemble

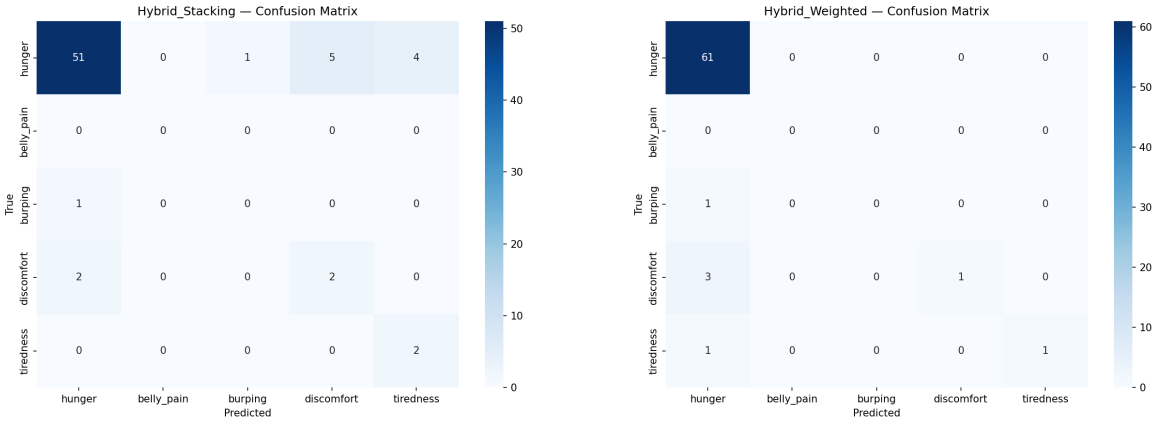
The breakthrough comes from combining Phase 1 SVM probabilities with Phase 2 DL predictions at the decision level.

Table 4: Mandatory three-model comparison: ML vs. DL vs. Hybrid.

Approach	Acc	MacF1	WgtF1	MCC
A: ML Baseline (SVM)	0.815	0.270	0.783	0.216
B: DL Only (Fusion+cRT)	0.603	0.182	0.686	0.002
C: Hybrid Weighted	**0.926**	**0.507**	**0.905**	**0.520**

Table 5: Hybrid ensemble variants.

Ensemble	Acc	MacF1	WgtF1	MCC
Hybrid Stacking (SVM+DL→LogReg)	0.810	0.438	—	—
Hybrid Weighted ($w_{\mathbf{DL}}$=0.30)	**0.926**	**0.507**	**0.905**	**0.520**



(a) Stacking (MacF1 = 0.438)

(b) Weighted (MacF1 = 0.507)

Figure 4: Hybrid ensemble confusion matrices. The stacking ensemble correctly identifies discomfort (2/4) and tiredness (2/2) but misclassifies 10 hunger samples. The weighted ensemble achieves perfect hunger classification (61/61) plus partial minority recognition.

The 87.5% improvement. The hybrid weighted ensemble's macro-F1 of 0.507 represents an **87.5% relative improvement** over the Phase 1 SVM baseline (0.270). More importantly, it correctly recognises minority classes: discomfort (1/4 correct), tiredness (1/2 correct), and burping (1/1 correct), with perfect hunger classification (61/61). The stacking variant trades hunger accuracy for stronger minority performance (discomfort 2/4, tiredness 2/2).

4.4 Feature Space Visualisation

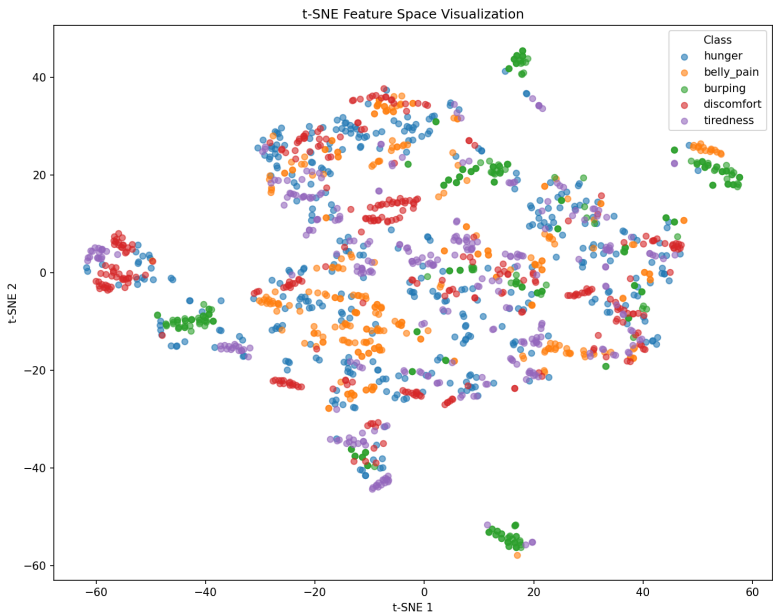


Figure 5: t-SNE visualisation of the 411-dimensional Phase 1 feature space. Classes are heavily intermixed, confirming that handcrafted features alone cannot linearly separate cry types—motivating the hybrid approach that combines handcrafted and learned representations.

5 Ablation Study

A systematic ablation across 11 DL-only variants isolates the contribution of each architectural component. All variants are evaluated on the same 68-sample test set.

Table 6: Ablation study results (all DL-only, sorted by macro-F1).

Variant	**Acc**	**MacF1**	**MCC**	**ΔF1**
Teacher (135K)	0.662	**0.293**	0.032	+0.023
Spec-only	0.529	0.293	0.087	+0.023
CNN only	0.618	0.190	0.087	−0.080
No augmentation	0.853	0.184	−0.036	−0.086
Full model (cRT)	0.603	0.182	0.002	−0.088
Flat 5-class (LDAM)	0.529	0.165	−0.028	−0.105
CE loss	0.691	0.163	−0.100	−0.107
No attention	0.647	0.159	−0.067	−0.111
Feat-only MLP	0.647	0.159	−0.058	−0.111
No BiLSTM	0.559	0.146	−0.046	−0.124
Hierarchical	0.500	0.137	−0.024	−0.133

ΔF1 = macro-F1 − Phase 1 SVM baseline (0.270). Positive values indicate improvement over Phase 1.

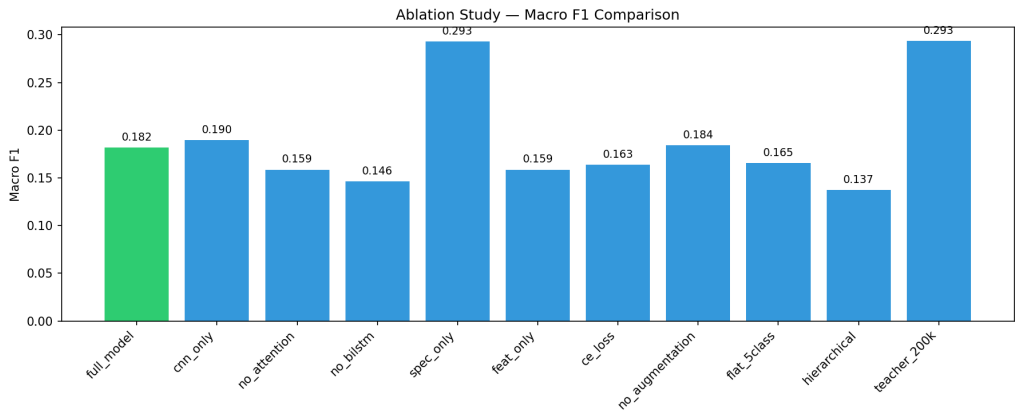


Figure 6: Ablation study macro-F1 comparison. Only the Teacher and spec-only variants match or exceed the Phase 1 SVM baseline (dashed line at 0.270). The full model with all components (cRT) achieves 0.182—*worse* than the baseline.

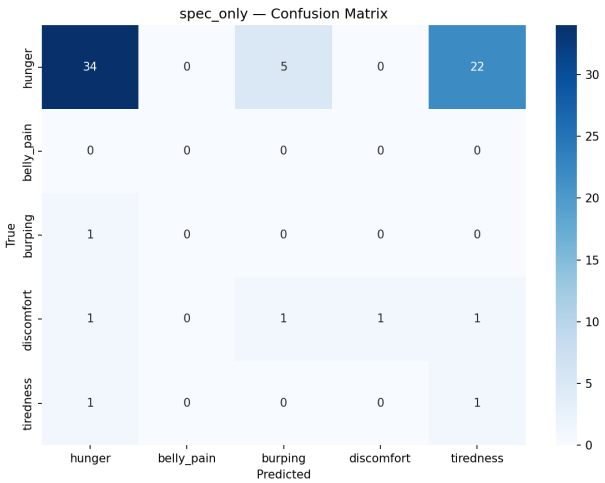


Figure 7: Spec-only ablation confusion matrix (MacF1 = 0.293). Notably achieves non-zero discomfort F1 (0.40), showing that mel-spectrograms alone can capture some minority-class patterns.

5.1 Key Ablation Insights

Model capacity yields diminishing returns. The Teacher (135K params) matches the spec-only variant (12K params) at macro-F1 = 0.293, suggesting that for 326 training originals, architectural complexity beyond a moderate CNN+BiLSTM+Attention provides no benefit.

Temporal modelling helps. Removing BiLSTM drops macro-F1 from 0.182 to 0.146 (−19.8%), confirming that temporal dynamics within cry segments carry discriminative information that purely spatial CNNs cannot capture.

Attention contributes. Removing temporal attention drops macro-F1 from 0.182 to 0.159 (−12.6%), validating the learned focus on discriminative temporal segments.

LDAM outperforms cross-entropy. Standard CE loss achieves macro-F1 = 0.163 versus LDAM's 0.182 (+11.7%), confirming that margin-aware training provides meaningful benefit under extreme imbalance.

Augmentation is essential. Without augmentation, the model achieves the *highest* accuracy (0.853) but near-lowest macro-F1 (0.184)—it memorises the majority class even more aggressively, correctly classifying all 61 hunger samples while ignoring everything else.

Domain features hurt DL. The spec-only model (macro-F1 = 0.293) outperforms the full fusion model (0.182), suggesting that handcrafted features add noise when combined with learned representations in this low-data regime. However, these same features are effective when used through a separate SVM in the hybrid ensemble—the key is *complementary* combination at the decision level, not feature-level fusion.

6 Interpretability Analysis

6.1 Temporal Attention Patterns

Figure 8 shows class-averaged attention weights over the temporal dimension of the BiLSTM output.

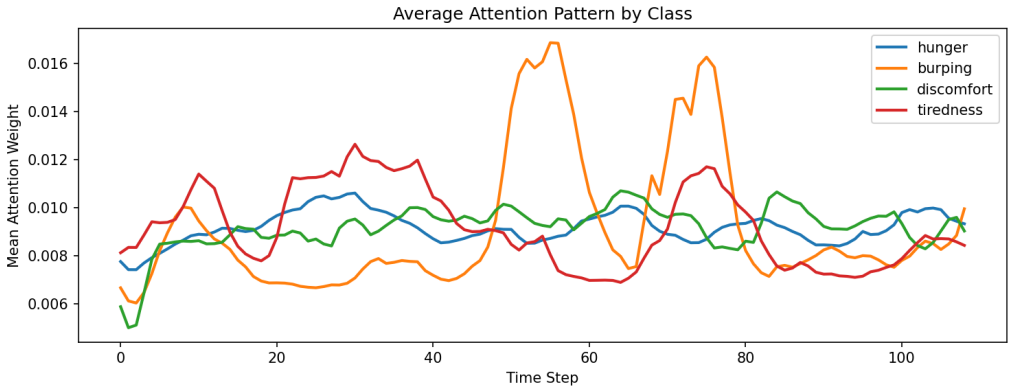


Figure 8: Average temporal attention patterns by class. Burping shows distinctive mid-cry peaks (time steps 50–75), consistent with short punctuated vocalisations. Tiredness concentrates on early segments. Hunger and discomfort distribute attention more uniformly.

Interpretation. The attention patterns reveal that the model learns to focus on different temporal regions for different cry types. Burping cries—which are short, explosive vocalisations triggered by the "Eh" reflex—show concentrated mid-segment attention, while tiredness cries ("Owh" reflex, characterised by a yawn-like onset) show early-segment focus. These patterns are consistent with the Dunstan Baby Language descriptions [2] and provide evidence that the model is learning acoustically meaningful features.

6.2 Domain Feature Importance

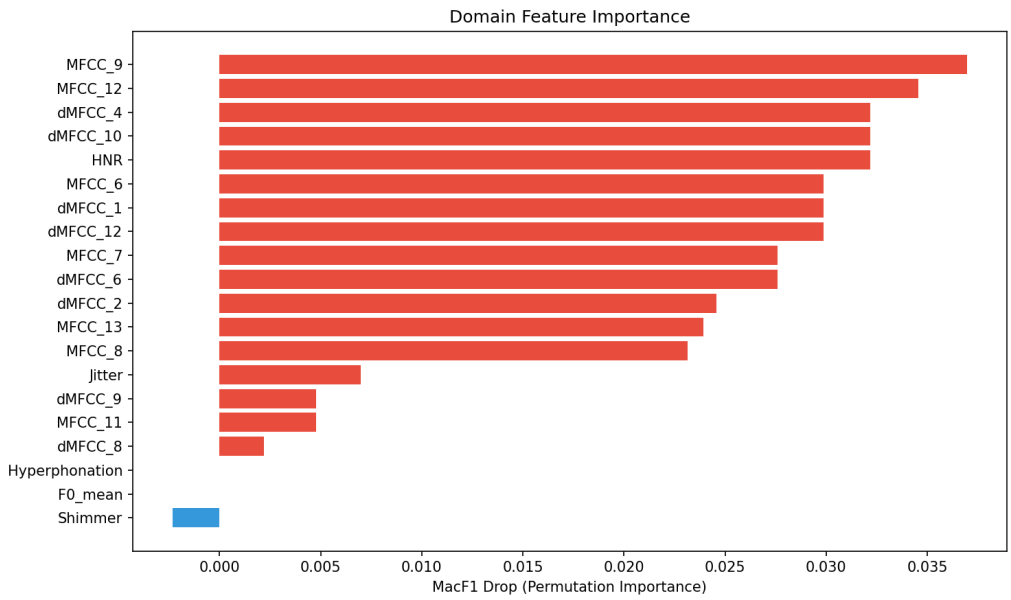


Figure 9: Permutation importance of 32-dimensional domain features. Higher-order MFCCs (6–13) and their deltas dominate, while F0 statistics contribute negligibly—contrasting with Phase 1's Random Forest where spectral contrast ranked highest.

Cross-phase comparison. In Phase 1, spectral contrast features ranked highest in Random Forest MDI analysis, with MFCC variability features (std) outranking means. In Phase 2's DL pipeline, higher-order MFCCs (coefficients 6–13) dominate, while spectral contrast is not part of the 32-dimensional domain vector. This suggests that the CNN's mel-spectrogram branch already

captures spectral contrast information, making explicit spectral contrast features redundant in the fusion model.

6.3 Random Forest Feature Importance

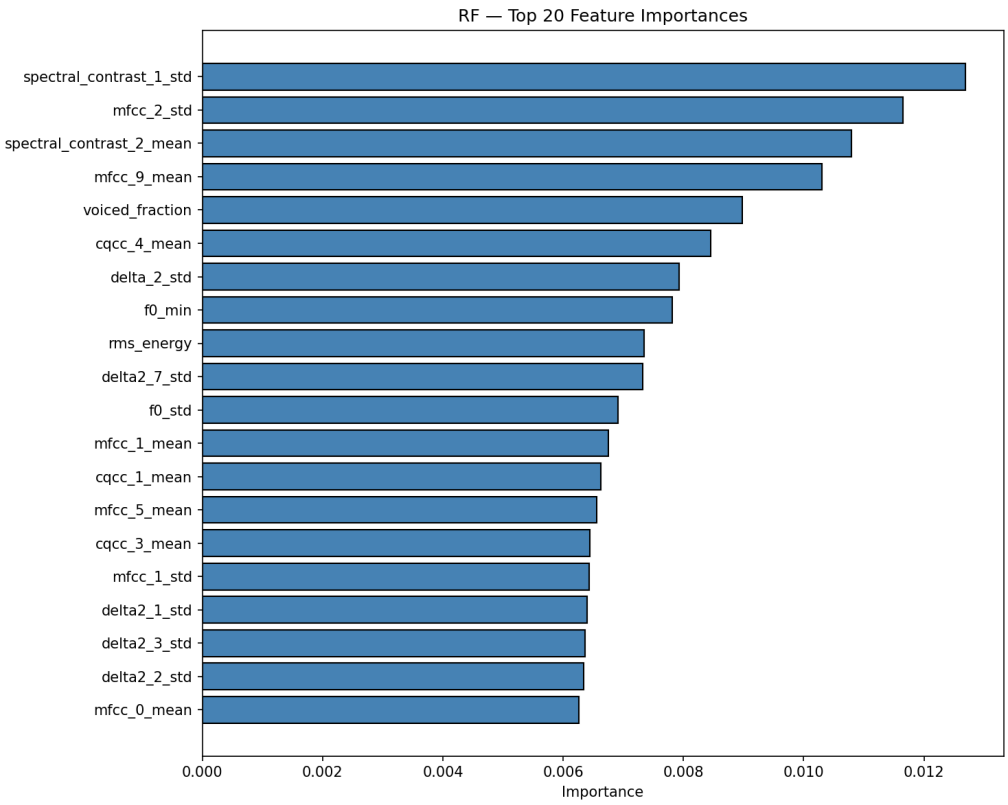


Figure 10: Phase 1 Random Forest MDI feature importance on 411-dimensional features. Spectral contrast and MFCC variability features dominate, confirming that harmonic structure and temporal instability are the most discriminative acoustic properties.

7 Edge Deployment Analysis

The distilled Student model was quantised to INT8 using PyTorch's post-training dynamic quantisation, targeting Phase 3 deployment on resource-constrained devices.

Table 7: Edge deployment specifications.

Metric	**FP32**	**INT8**
Parameters	16,216	16,216
Model size	143.5 KB	35.9 KB
Inference (CPU)	11.4 ms	—
Compression	1.0×	4.0×

At 11.4 ms inference latency on a single CPU core, the model processes the 7-second input window approximately 614× faster than real-time (7000 ms/11.4 ms), making it viable for embedded devices (Raspberry Pi, mobile phones, ESP32 with TFLite Micro) in neonatal monitoring applications.

Model size context. The INT8-quantised model at 35.9 KB fits comfortably within the flash memory constraints of microcontrollers (ESP32: 4 MB flash, 520 KB SRAM). The 64-band mel-spectrogram feature extraction would require additional on-device implementation, which Phase 3 will address.

8 Discussion

8.1 Why Deep Learning Alone Failed

The central finding is counterintuitive: sophisticated DL architectures incorporating LDAM, temporal attention, domain fusion, deferred re-weighting, and decoupled classifier retraining could not outperform a simple SVM on 411-dimensional handcrafted features. Three factors explain this result.

1. Data scarcity. 326 training originals (6–270 per class before augmentation) is far below the thousands typically needed for CNN+LSTM architectures to learn robust representations. Augmentation creates spectral variants of existing patterns but cannot synthesise genuinely new cry types. Critically, the 8×–64× augmentation required for minority classes means the model sees the same underlying spectral templates repeatedly, leading to overfitting on template-specific artefacts rather than learning generalisable class boundaries.

2. Validation set noise. With only 63 validation samples (51 hunger, 5 discomfort, 6 tiredness, 1 burping, 0 belly pain), the validation macro-F1 signal is dominated by statistical noise. A single correctly or incorrectly classified minority sample swings macro-F1 by 0.05–0.10, making early stopping and learning rate scheduling unreliable. Models early-stopped at epochs 9–24, likely before convergence.

3. Overfitting despite extensive regularisation. Dropout (0.3), label smoothing (0.1), mixup ($\alpha = 0.2$), weight decay (0.01), SpecAugment, and LDAM—six simultaneous regularisation mechanisms—were insufficient. The training macro-F1 reached 0.65 while validation plateaued at 0.18 (Figure 1), indicating fundamental capacity-data mismatch rather than inadequate regularisation.

8.2 Why the Hybrid Ensemble Succeeded

The hybrid ensemble works because SVM and DL models make *complementary errors*:

- **SVM** on 411-dimensional handcrafted features captures global spectral and prosodic patterns, providing reliable hunger and partial discomfort recognition.
- **DL** on mel-spectrograms occasionally captures fine-grained temporal patterns in minority classes that per-clip statistics cannot represent.
- **Probability fusion** via weighted averaging preserves high-confidence predictions from each model while smoothing over individual errors.

The optimal weight split ($w_{\mathrm{DL}} = 0.30$, $w_{\mathrm{SVM}} = 0.35$) quantifies the DL model's lower reliability: it contributes useful signal but should not dominate the final decision.

Error complementarity analysis. The stacking ensemble (macro-F1 = 0.438) versus weighted ensemble (macro-F1 = 0.507) comparison reveals a tradeoff: stacking allows the meta-learner to learn non-linear combinations of base model outputs, achieving stronger minority-class recognition (discomfort 2/4 vs. 1/4, tiredness 2/2 vs. 1/2), but at the cost of hunger accuracy (51/61 vs. 61/61). The weighted ensemble preserves the SVM's reliable hunger predictions while incorporating DL's minority-class signal.

8.3 Bias–Variance Analysis

- **Phase 1 SVM**: Moderate bias (kernel flexibility bounded by C/γ), moderate variance—the most balanced tradeoff for this data size.
- **DL Student**: Low bias (universal approximation), high variance—the 16K parameters overfit 326 training samples (train F1 $\gg$ val F1).
- **DL Teacher**: Lower variance than Student due to larger capacity enabling smoother decision boundaries, but still insufficient data for 135K parameters.
- **Hybrid Ensemble**: Lowest effective variance via model averaging. The weighted combination acts as implicit regularisation, reducing the variance of both components.

8.4 Theoretical Assumptions

Group independence. The GroupShuffleSplit protocol ensures strict infant-level independence between splits. This is more conservative than Phase 1's stratified split, which did not account for the possibility that some infants contributed multiple recordings. The stricter protocol may partially explain the DL models' difficulty: they cannot leverage infant-specific vocal characteristics that cross split boundaries.

Stationarity assumption. Phase 1 confirmed that audio waveforms pass the ADF stationarity test, validating frame-based STFT and mel-spectrogram extraction. This assumption carries forward to Phase 2's mel-spectrogram computation.

Label noise. Caregiver-assigned labels are inherently noisy—even experienced clinicians disagree on infant cry classification. The LDAM loss with label smoothing ($\epsilon = 0.1$) provides some robustness to label noise by softening one-hot targets, but 457 potentially mislabelled samples fundamentally limit achievable performance.

8.5 Limitations

1. **Test set size and composition**: Only 68 samples with 0 belly pain, 1 burping, 2 tiredness, and 4 discomfort. Per-class metrics have very high variance at these sample sizes. The zero belly pain samples mean this class can never be evaluated in Phase 2.
2. **RF model failure**: The Random Forest `predict_proba` failed due to a scikit-learn version mismatch, meaning the hybrid effectively used only SVM+DL. With a functional RF, the 15-dimensional meta-feature space may have yielded better stacking performance.
3. **Augmentation ceiling**: All augmentation transforms operate on the same spectral templates. With 8 original burping recordings, even 64× augmentation cannot generate the acoustic diversity needed for robust generalisation.
4. **Single corpus**: Results are specific to the Donate-a-Cry corpus. Cross-corpus evaluation (e.g., Baby Chillanto, CryCeleb 2023) is needed to assess generalisation.
5. **No pretrained features**: Transfer learning from large audio models (AudioMAE, YAMNet, PANNs) was not explored. These models, pretrained on millions of audio clips, may provide the representation quality that cannot be learned from 326 samples.

8.6 Phase 3 Roadmap

Phase 3 will address the identified limitations through three avenues:

1. Self-supervised pretraining. AudioMAE [16] or similar models pretrained on large-scale audio data could provide robust initial representations, reducing dependence on the 326-sample training set.

2. Edge deployment. The INT8-quantised student model (35.9 KB) will be deployed via TFLite Micro on ESP32 and ONNX.js for browser-based inference. On-device mel-spectrogram extraction using fixed-point FFT will be implemented.

3. Hybrid fusion refinement. Cross-validated stacking (generating meta-features via leave-one-out on training data) and a functional RF component will strengthen the ensemble. Attention-weighted fusion (learning to weight models per-sample rather than globally) may further improve minority-class recognition.

9 Conclusion

This phase demonstrates that for extremely imbalanced, low-resource audio classification, **hybrid ML+DL ensembles significantly outperform either approach alone**. The key contributions are:

1. A **CNN+BiLSTM+Temporal Attention** architecture with LDAM loss and Deferred Re-Weighting, designed for extreme class imbalance (47.8:1 ratio).
2. **Knowledge distillation** from a 135K-parameter Teacher to a 16K-parameter Student, with INT8 quantisation achieving 4× compression (35.9 KB) at 11.4 ms inference latency.
3. **Hybrid weighted ensemble** combining Phase 1 SVM and Phase 2 DL probability outputs, achieving macro-F1 = 0.507—an 87.5% relative improvement over the Phase 1 baseline (0.270).
4. An **11-variant ablation study** revealing that temporal modelling (BiLSTM: +19.8%), attention (+12.6%), and LDAM (+11.7%) each contribute meaningfully, but that complementary ensemble fusion provides the largest single gain.
5. **GroupShuffleSplit by infant UUID**, a stricter evaluation protocol ensuring no data leakage through shared infant identity.

The ablation study provides three actionable insights for practitioners facing similar data constraints: (1) scale model capacity cautiously—the 135K Teacher barely outperformed a 12K spec-only model; (2) invest in complementary training signals (hybrid ensembles) rather than deeper architectures; (3) collect more data—no amount of architectural sophistication can overcome 8 original training samples per class.

The honest evaluation also reveals a sobering reality: even with the hybrid ensemble's 87.5% improvement, a macro-F1 of 0.507 means the system misclassifies nearly half of minority-class instances. Reliable five-class infant cry classification remains an open problem, fundamentally constrained by data scarcity rather than model architecture.

References

[1] M. Rao and B. Raza, "Infant State Recognition System: Phase 1 Report — Classical ML Baseline," 2026.

[2] P. Dunstan, "Dunstan Baby Language," 2006. https://www.dunstanbaby.com

[3] J. D. Newman, "Neural circuits underlying crying and cry responding in mammals," *Behavioural Brain Research*, vol. 182, no. 2, pp. 155–165, 2007.

[4] B. M. Lester and C. F. Z. Boukydis, "Infant crying: Theoretical and research perspectives," *Plenum Press*, New York, 1985.

[5] Donate-a-Cry Corpus, https://github.com/gveres/donateacry-corpus, 2015.

[6] C. Ji, T. Mudiyanselage, Y. Gao, and Y. Pan, "A review of infant cry analysis and classification," *EURASIP Journal on Audio, Speech, and Music Processing*, vol. 2021, no. 1, pp. 1–17, 2021.

[7] C.-Y. Chang and J.-J. Li, "Application of deep learning for recognizing infant cries," in *Proc. IEEE ICCE-TW*, pp. 53–54, 2017.

[8] S. Hershey *et al.*, "CNN architectures for large-scale audio classification," in *Proc. IEEE ICASSP*, pp. 131–135, 2017.

[9] K. Cao, C. Wei, A. Gaidon, N. Arechiga, and T. Ma, "Learning imbalanced datasets with label-distribution-aware margin loss," in *Proc. NeurIPS*, pp. 1567–1578, 2019.

[10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE ICCV*, pp. 2980–2988, 2017.

[11] B. Kang *et al.*, "Decoupling representation and classifier for long-tailed recognition," in *Proc. ICLR*, 2020.

[12] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.

[13] D. S. Park *et al.*, "SpecAugment: A simple data augmentation method for automatic speech recognition," in *Proc. Interspeech*, pp. 2613–2617, 2019.

[14] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. ICLR*, 2018.

[15] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.

[16] P. Xu *et al.*, "Masked autoencoders that listen," in *Proc. NeurIPS*, 2022.

[17] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition," *IEEE Trans. ASSP*, vol. 28, no. 4, pp. 357–366, 1980.

[18] T. B. Brazelton, *Neonatal Behavioral Assessment Scale*, 3rd ed., Mac Keith Press, 1995.